

Introduction to Data Science

Assignment 1

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The Tableau Public:

https://public.tableau.com/app/profile/paria.pasehvarz/viz/HW1_17436096033690/AirbnbHotspots?publish=yes

Task 2: Tableau

In this task, we will utilize Tableau to analyze Airbnb data and create a data-driven story through multiple interactive dashboards. The objective is to build a well-structured narrative using several dashboards, each focusing on different aspects of the data.

Dataset Details

1. Airbnb_Listings.xls - contains information about Airbnb listings, including:

- Neighborhood – The specific area where the listing is located.
- Price – Cost per night for each listing.
- Availability 365 – Number of days a listing is available for booking per year.
- Calculated Host Listings Count – The total number of listings a host has.
- Host Id – Unique identifier for the host.
- Host Name – Name of the host.
- Id – Unique identifier for the listing.
- Last Review – Date of the most recent review for the listing.
- Minimum Nights – Minimum number of nights required per booking.
- Name – Name of the Airbnb listing.

- Number Of Reviews – Total number of reviews received for the listing.
- Reviews Per Month – Average number of reviews per month.
- Room Type – Type of room (e.g., Entire home/apt, Private room, Shared room).

2. Neighborhood_Locations.xlsx – Provides latitude, longitude, and neighborhood group classifications for mapping.

- Neighborhood: The specific area where the Airbnb listing is located.
- Neighborhood Group: A broader classification of neighborhoods.
- Latitude: The geographical latitude of the listing.
- Longitude: The geographical longitude of the listing.

1. Data Preparation

1. First, we imported datasets into the Tableau and joined them together by field neighborhood:

Airbnb_Analysis

airbnb is made of 2 tables. ⓘ

airbnb —  — Neighbourhood

Join

 Inner  Left  Right  Full Outer

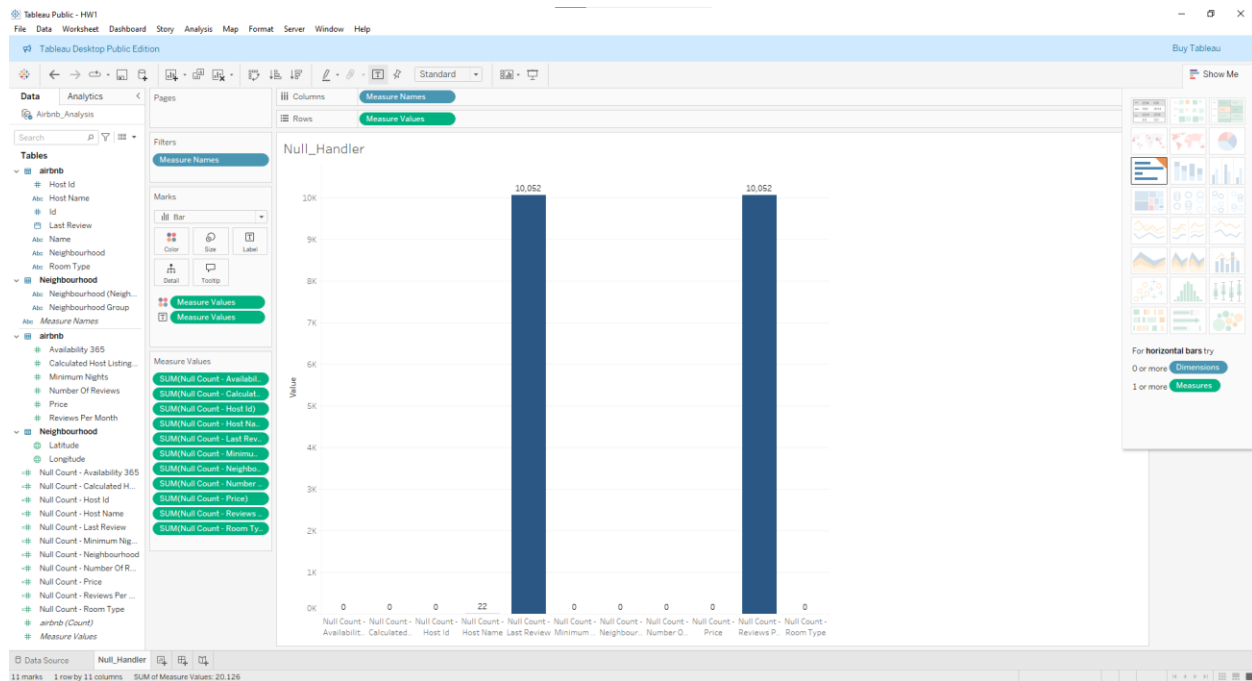
Data Source		Neighbourhood	
Neighbourhood	=	Neighbourhood ...	
Add new join clause			

2. In another sheet named “Null_Handler”, we handle null values in datasets. First, we create some calculated filters for each column to identify rows which has a null value.

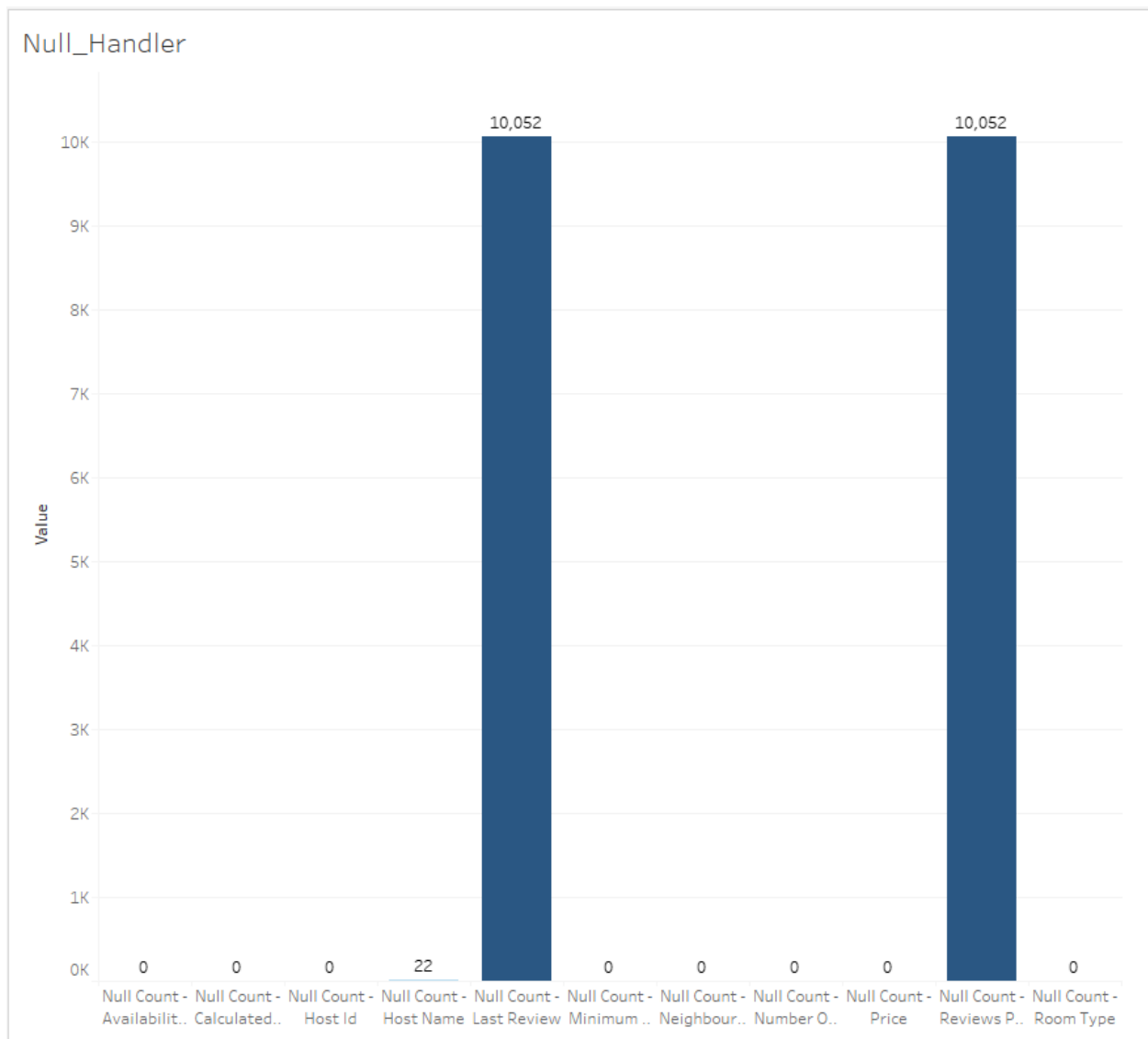
The formula for it, for example for Host ID column is as follows:

IF ISNULL([Host Name]) THEN 1 ELSE 0 END

Now we use these filters and plot a bar chart which demonstrates number of null values in each column:



And a closer view:



We decided to clean null values and replace them, so:

- For Last Review, we replaced it we replaced it with a default date.

```
IF ISNULL([Last Review]) THEN DATE("2000-01-01") ELSE [Last Review] END
```

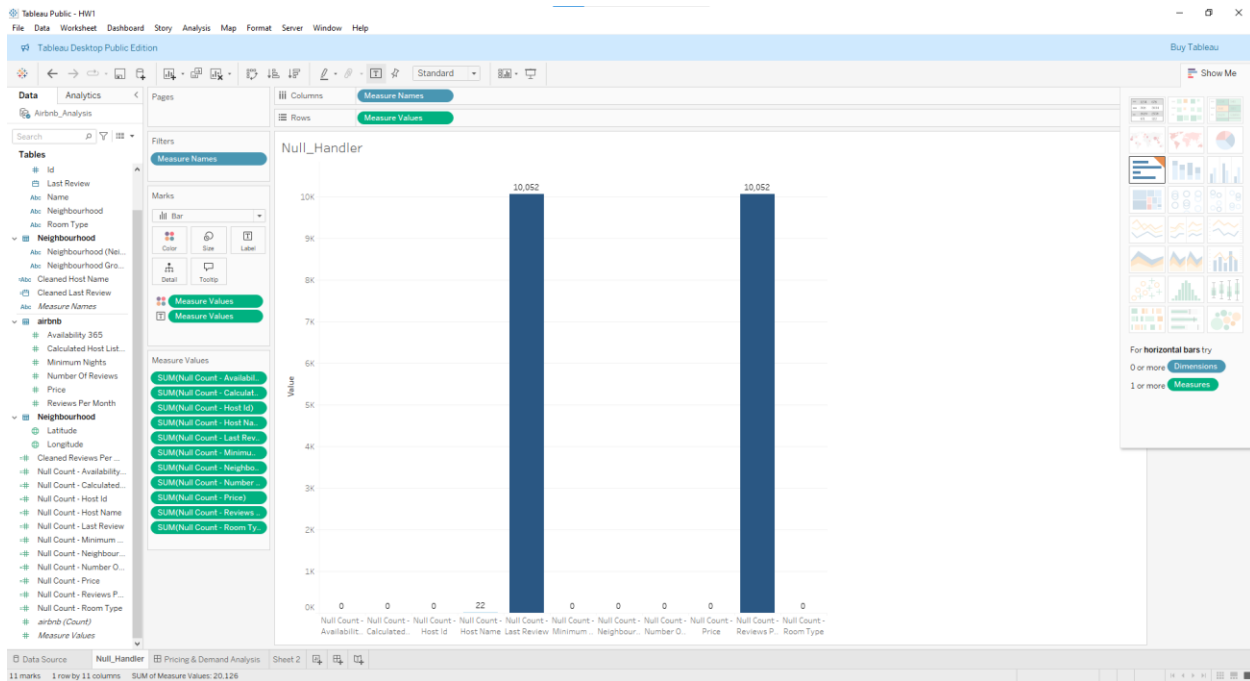
- For Reviews per month, we replaced it with 0.

IF ISNULL([Reviews Per Month]) THEN 0 ELSE [Reviews Per Month]
END

- For Host Name, we replaced it with Unknown host:

IF ISNULL([Last Review]) THEN "No Reviews" ELSE STR([Last
Review]) END

Finally, Nul_Handler sheet looks like this:



2. Building Dashboards

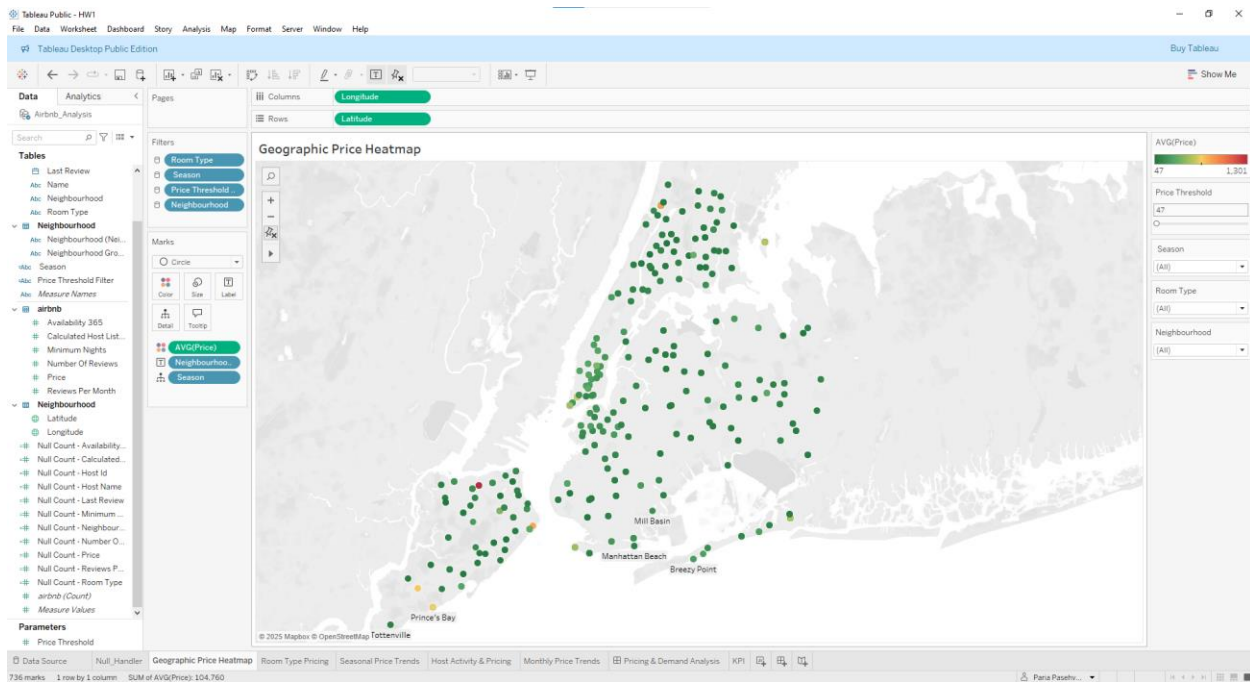
Dashboard 1: How Much Does It Cost to Stay?

We analyze how Airbnb prices vary based on location, seasonality, and room type.

Sheet 1: Geographic Price Heatmap

First, we create a Geographic Price Heatmap which show which neighborhoods are expensive or affordable.

We used longitude and latitude info to inspect about prices in different neighborhoods:



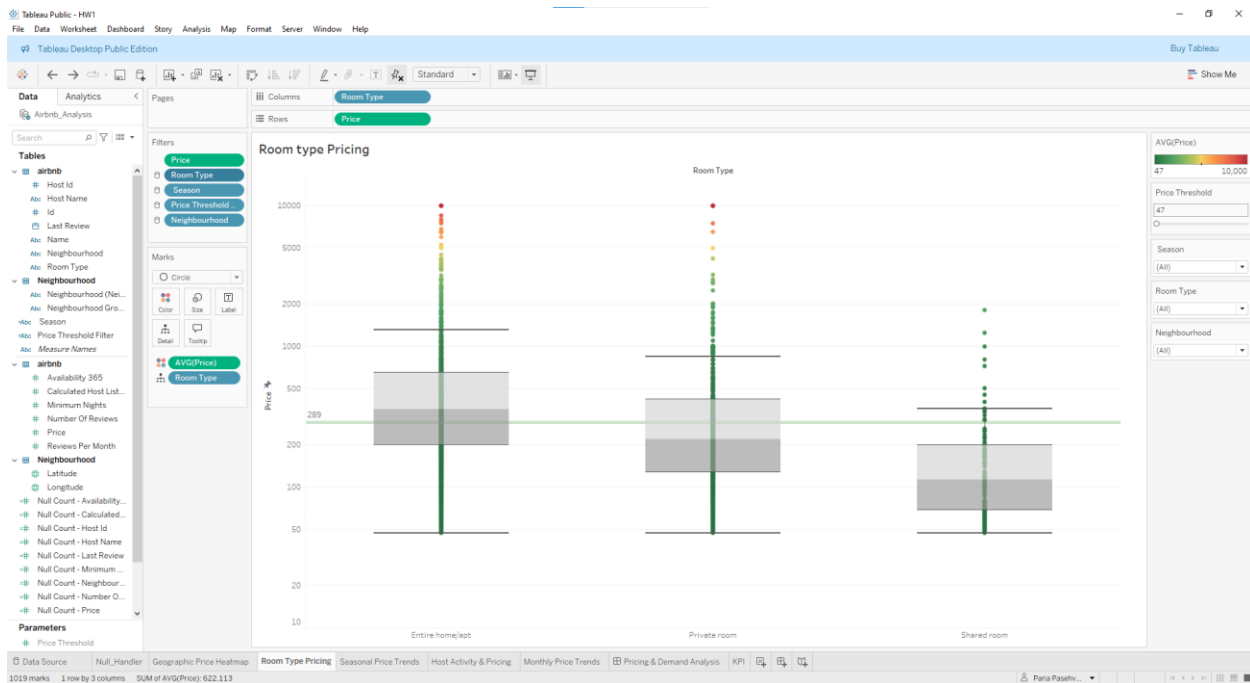
The colors used: red represents expensive areas, while green indicates cheaper ones. This allows us to observe how prices vary across different neighborhoods.

We added filters for room type, season and neighborhood, enabling users to explore specific areas or price ranges.

A new parameter, the price threshold, has also been introduced. When combined with the price threshold filter, it allows users to set a specific price limit and visualize the changes on the map.

Sheet 2: Room Type Pricing Comparison

This visualization provides a comparative analysis of price distributions across various room types, specifically focusing on Entire Homes, Private Rooms, and Shared Rooms. By examining the spread and central tendencies of prices within each category, we can better understand the pricing dynamics and how they differ based on the type of accommodation. This comparison allows us to identify patterns such as which room type generally commands higher prices, how much variation exists within each category, and whether certain room types are more affordable or premium in the overall market. Filters and thresholds are the same with previous sheets.



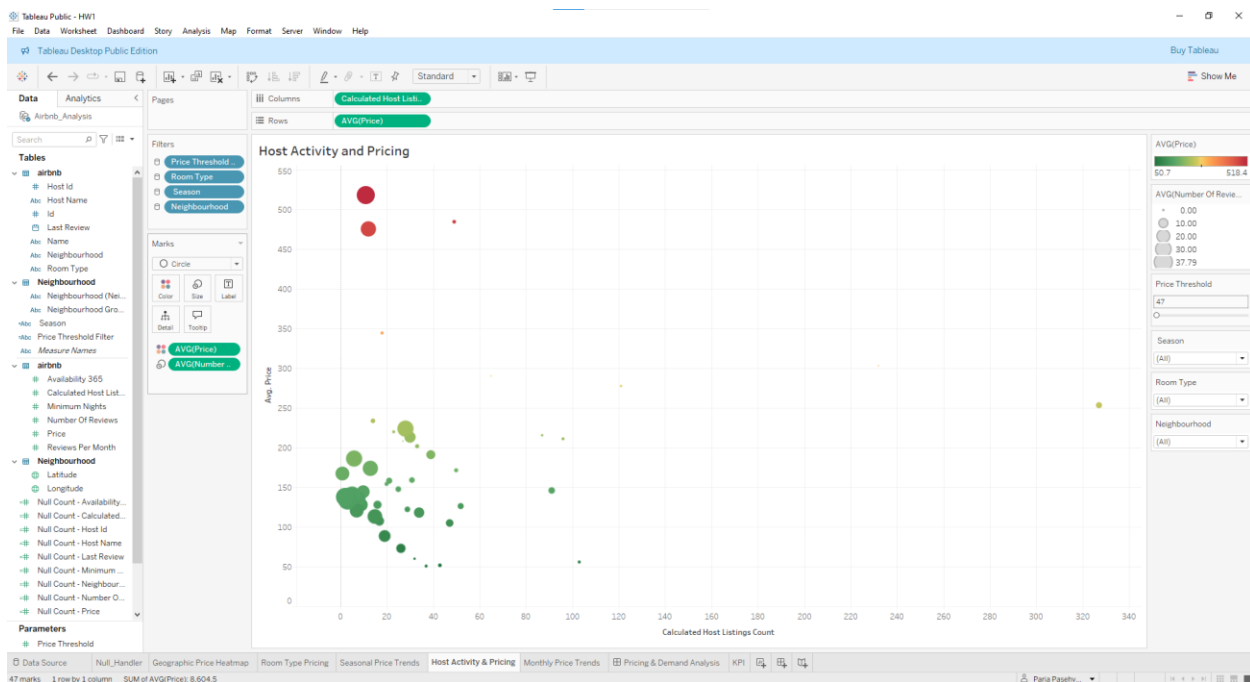
Sheet 3: Seasonal Price Trends

This visualization presents a comparative analysis of average prices across different seasons over a span of several years. By examining these seasonal averages year by year, we can gain insights into the pricing trends and fluctuations that occur throughout the calendar. The visual format makes it easier to identify notable patterns, such as consistent increases or decreases, as well as significant spikes or drops in price. In particular, it highlights the seasons and years where peak prices occurred, helping us understand potential market dynamics, external factors, or seasonal demand shifts that may have contributed to those changes. Filters and thresholds are the same with previous sheets.



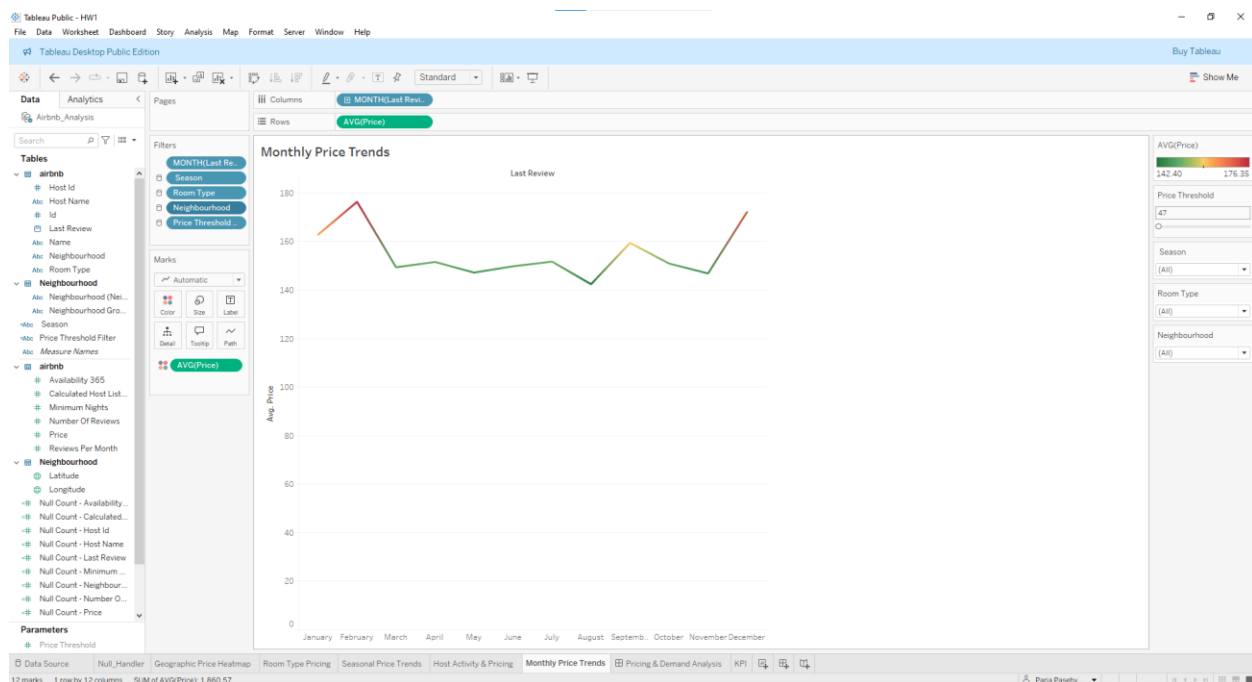
Sheet 4: Host Activity & Pricing

In this visualization, we first calculated the number of listings associated with each host, and then used that information to compute the average price per host. The scatter plot uses multiple visual elements to convey insights: the x and y axes represent the relevant metrics (e.g., listing count and price), while the bubble size indicates the average number of reviews each host receives. The color of each bubble reflects the average price, helping to visualize how prices vary among different hosts — with red tones representing more expensive listings, and green tones indicating more affordable ones. This multi-dimensional approach allows us to simultaneously analyze host activity, pricing, and popularity (as measured by reviews), and to spot trends or outliers across the hosting landscape. Filters and thresholds are the same with previous sheets.



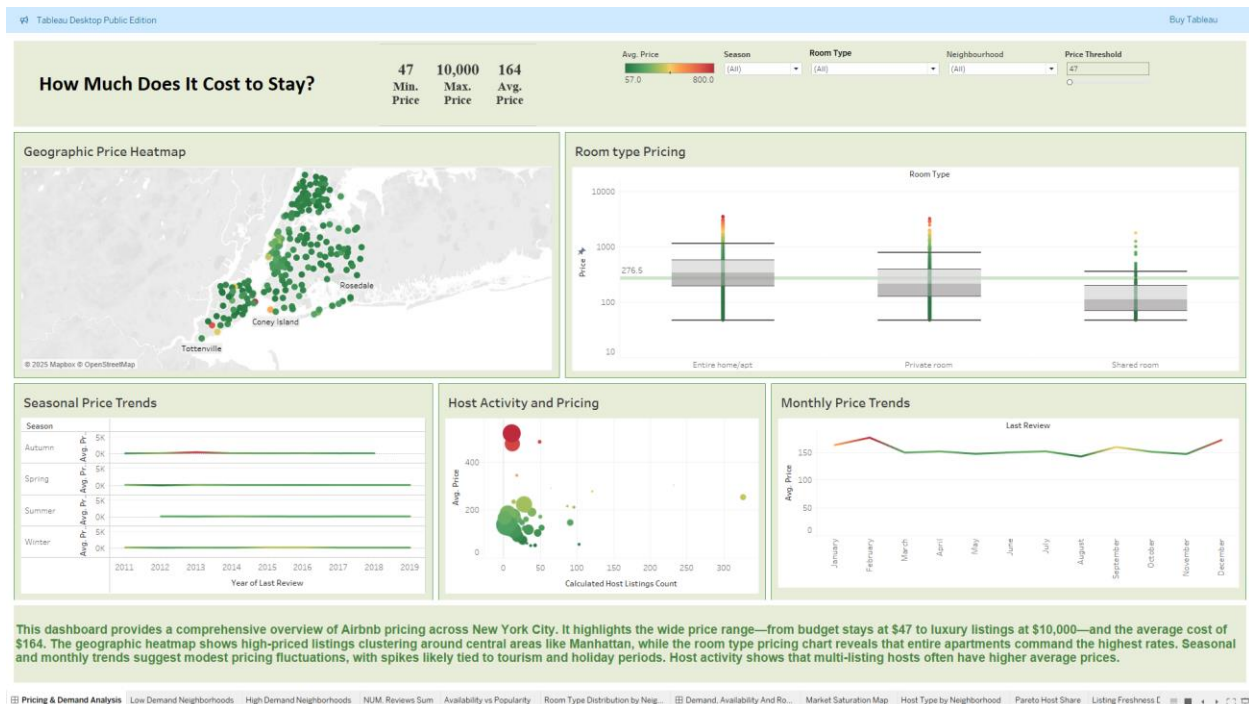
Sheet 5: Monthly Price Trends

In this visualization, we compare the average prices across the months of a calendar year to identify seasonal trends in pricing. Each month is represented with a color gradient that reflects the average price: higher prices are shown in red, while lower prices are depicted in green. This color-coded approach makes it easy to visually spot which months tend to be more expensive and which ones are more affordable, offering valuable insights into seasonal demand and pricing patterns over time. Filters and thresholds are the same with previous sheets.



Final Dashboard

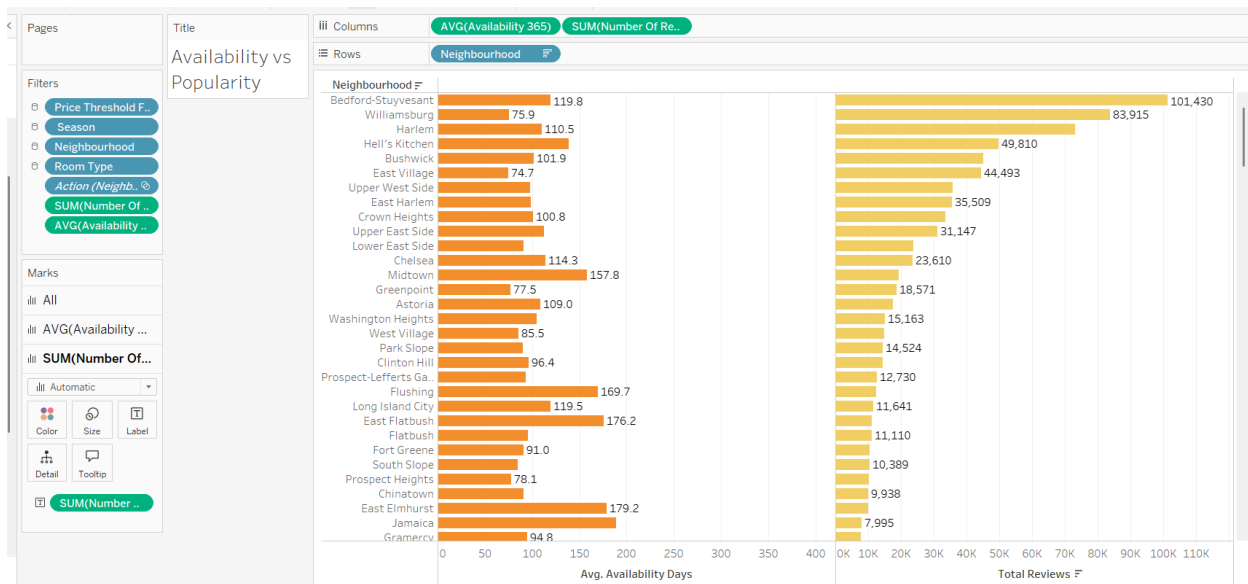
Finally, we combined all the individual sheets into a single, cohesive dashboard. This integrated view allows for a more comprehensive analysis by bringing together all key visualizations in one place. The dashboard includes interactive filters that are synchronized across all sheets, enabling users to explore the data dynamically based on selected criteria such as room type, location, or host activity. Additionally, key performance indicators (KPIs) are displayed at the top, providing a quick summary of important metrics — including the minimum price, maximum price, and average price — to give users immediate insight into the overall pricing landscape



Dashboard 2: Which NYC Neighborhood Is Right for You?

Sheet 1: Availability vs. Popularity

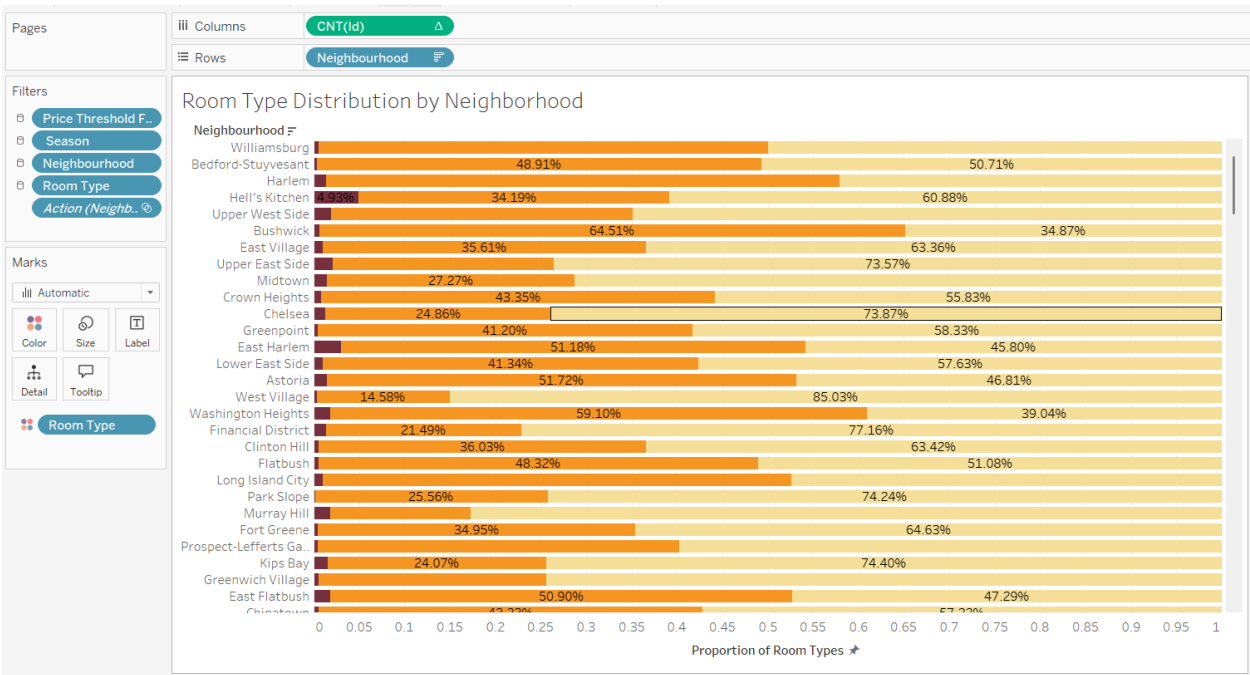
This sheet compares neighborhoods based on two key factors: average listing availability (left) and total number of reviews (right), which serves as a proxy for demand. Each row represents a neighborhood, and neighborhoods are sorted by total review count, placing high-demand areas at the top. To create this view, we used the average number of available days as the value and label for the left chart, and the total number of reviews as the value and label for the right chart. By examining both bars side by side, we can observe that many of the most popular neighborhoods tend to have lower average availability, while areas with higher availability often have lower demand. This suggests that although high-demand neighborhoods are attractive, they may offer fewer available days for booking—an important consideration for travelers seeking longer or more flexible stays.



Sheet 2: Availability vs. Popularity

This visualization presents the distribution of room types across different neighborhoods in New York City. Each horizontal bar represents a single neighborhood and is divided into color-coded segments corresponding to the proportion of each room type—such as entire home/apartment, private room, and shared room. The calculation is based on the percentage of distinct hosts offering each room type within that neighborhood, ensuring that the distribution reflects host diversity rather than just listing volume.

By examining this chart, users can easily compare neighborhoods based on the dominant types of accommodation available. For instance, a user looking specifically for an entire home or apartment can quickly identify which neighborhoods have a higher concentration of such listings. This helps users assess whether a particular neighborhood aligns with their accommodation preferences, providing a clear and practical tool for decision-making when selecting where to stay.

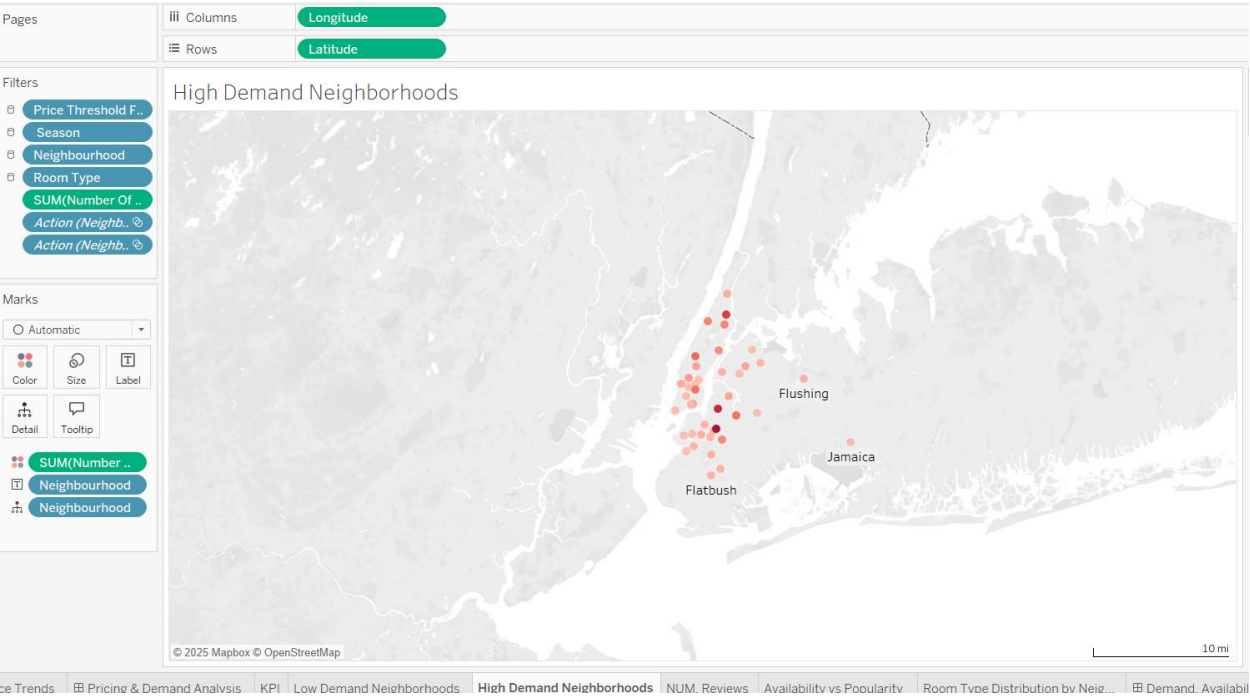
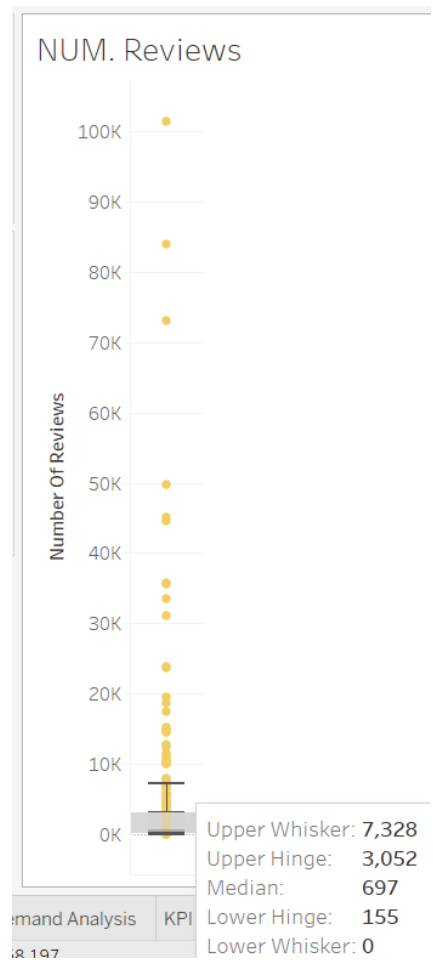


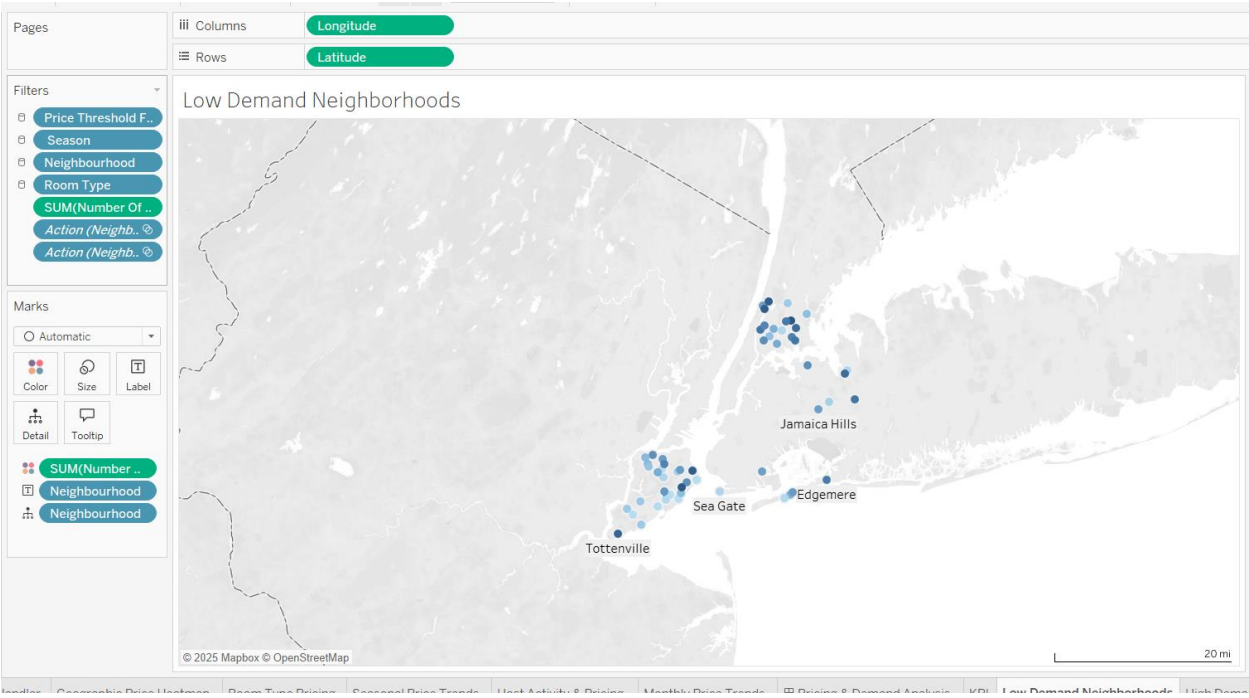
Sheet 3, 4: High and Low Demand Neighborhoods

In this map, we plot each neighborhood using its geographic coordinates (latitude and longitude). The color intensity of each point represents the total number of reviews, serving as an indicator of popularity. Each point is also labeled with the corresponding neighborhood name for clarity.

To provide more focused insights, we create two separate maps: one for high-demand neighborhoods and one for low-demand neighborhoods. The classification is based on a threshold derived from the box plot of total reviews. Specifically, neighborhoods below the first quartile (155 reviews) are categorized as low demand, while those above the third quartile (3,052 reviews) are considered high demand.

By applying these filters, we produce two distinct visualizations that highlight the spatial distribution of neighborhoods based on demand, allowing users to visually compare less popular areas with those that attract significantly more attention.





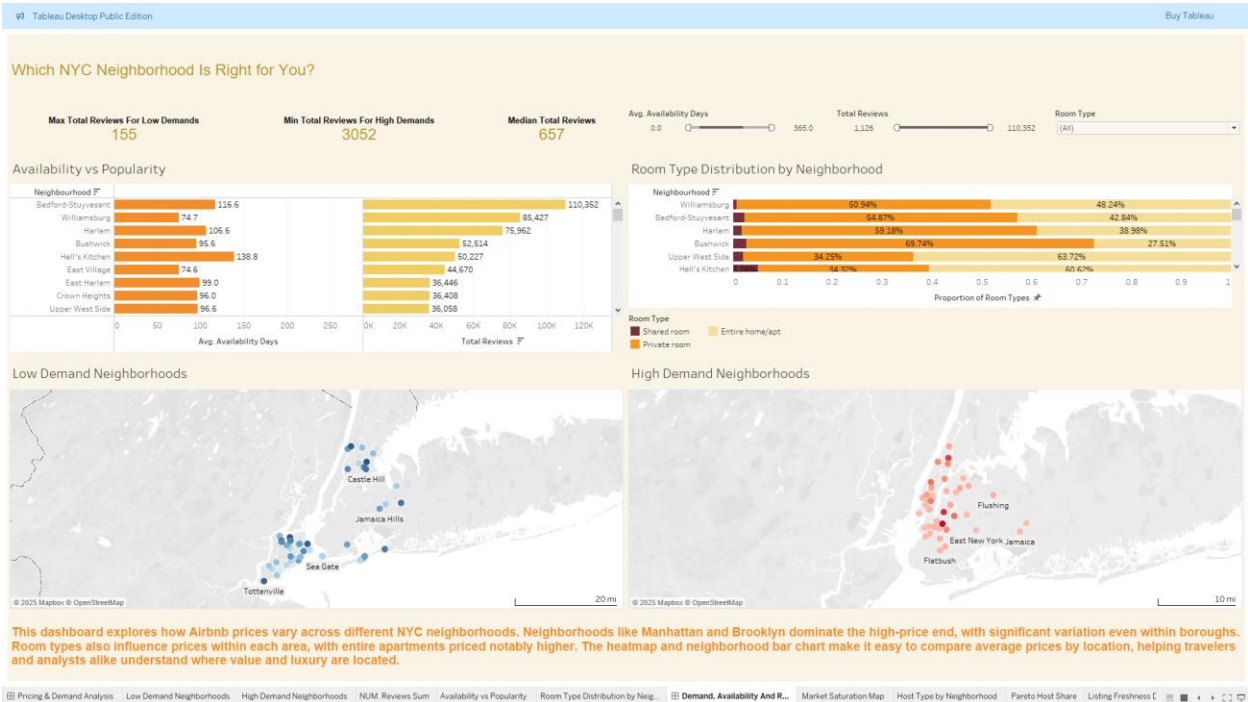
Final Dashboard

The final version of the dashboard provides a comprehensive overview of Airbnb activity across New York City neighborhoods, structured to guide users in identifying the most suitable area based on their accommodation needs. The charts are arranged in order of importance, beginning with the "Availability vs Popularity" chart, which is the most significant. This section highlights how frequently listings are available and how popular each neighborhood is based on the total number of reviews.

Following that, the "Room Type Distribution by Neighborhood" chart displays the proportion of shared rooms, private rooms, and entire homes within each area, offering users a clear view of what types of accommodations are predominant in different neighborhoods. Lastly, two geographic maps illustrate the spatial distribution of neighborhoods with either low or high demand, classified based on review volume.

To enhance user interaction and personalization, the top two charts include filters for room type, total reviews, and average availability days. These filters allow users to narrow down results according to their specific preferences. Additionally, the dashboard includes key summary statistics derived from the data: the low-demand threshold is set at 155 total reviews (first quartile), the high-demand threshold at 3,052 reviews (third quartile), and the median total reviews per neighborhood is 657. These figures provide immediate context and help users interpret the visualizations more effectively.

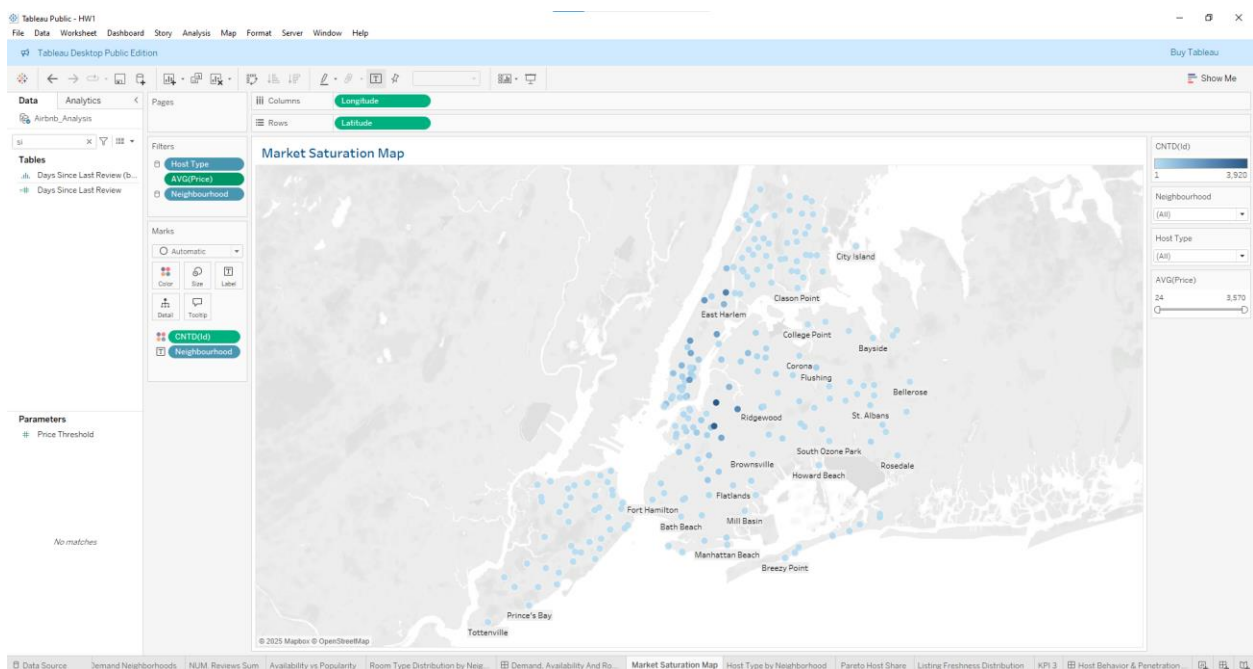
Overall, the dashboard combines clear visuals, interactive elements, and statistical context to support informed decision-making when selecting a neighborhood and room type in New York City.



Dashboard 2: Unlocking the Market: Host & Penetration Trends

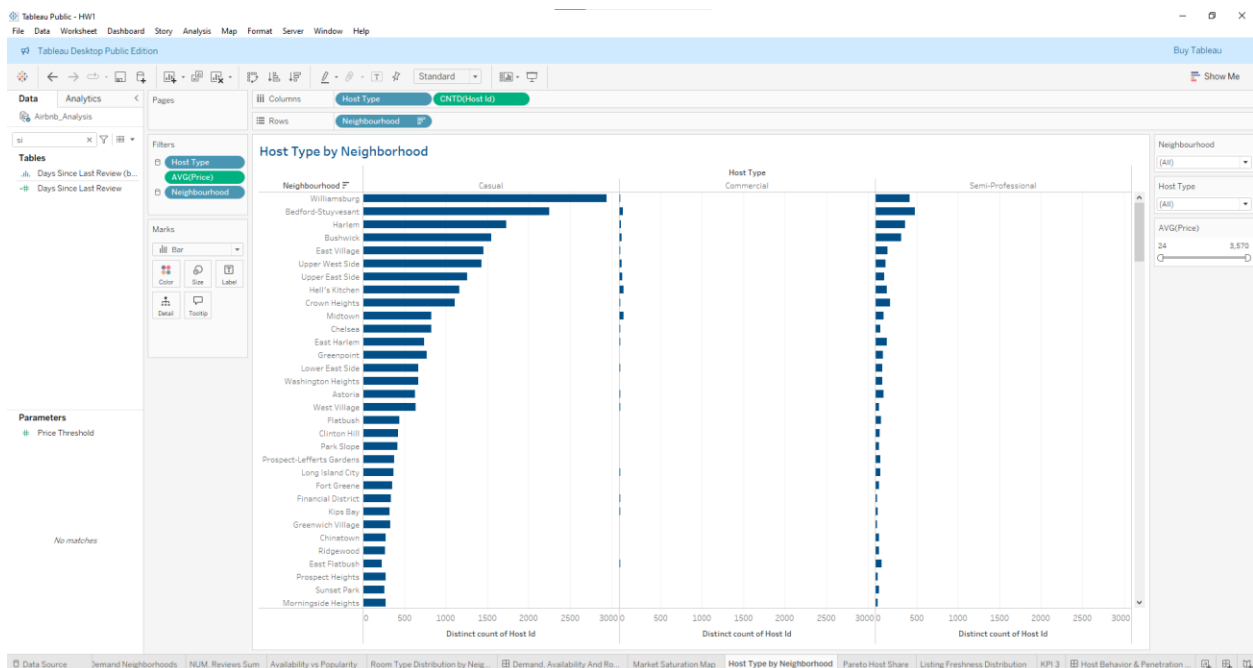
Sheet 1: Market Saturation Map

This visualization provides a visual heatmap representation of Airbnb listing density across New York City, offering a geographic perspective on market concentration. By plotting latitude and longitude on a map and adding neighborhood as a detail, the chart highlights how listings are distributed spatially across the boroughs. The use of a distinct count of listing IDs on the color scale creates a clear gradient—areas with a high volume of listings appear in darker shades, signaling greater saturation, while lighter areas indicate fewer listings and potentially untapped markets. This visualization is especially valuable for hosts, property managers, and market analysts, as it reveals competitive pressure in different regions. For example, highly saturated zones like parts of Manhattan or Brooklyn might suggest a need for strategic differentiation, while less saturated neighborhoods could present opportunities for new listings. Adjusting the color scale further enhances interpretability, allowing users to pinpoint clusters and compare saturation levels with ease. Overall, this step is essential for understanding spatial dynamics and guiding data-driven decisions in a competitive rental market.



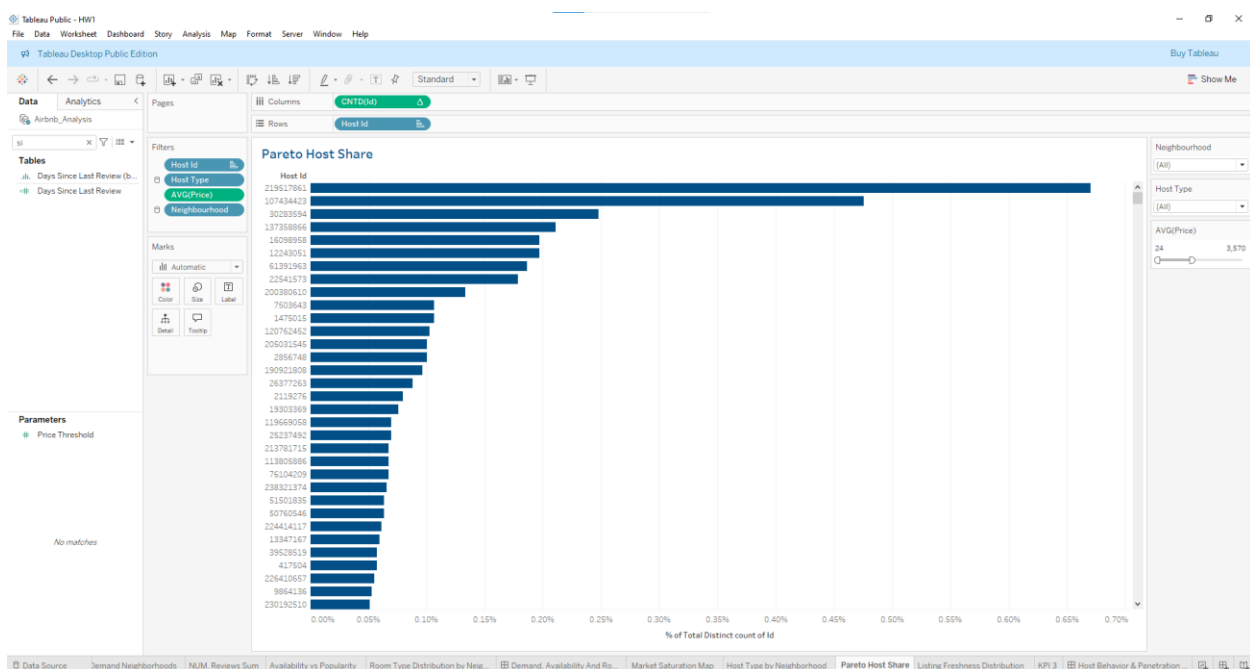
Sheet 2: Host Type by Neighborhood

This visualization offers a comparative view of how different types of Airbnb hosts are distributed across various neighborhoods in New York City. By placing neighborhoods along the rows and categorizing host types across the columns, this chart enables a clear breakdown of hosting behavior at the local level. The use of the distinct count of Host IDs as a measure ensures that the data reflects the number of unique hosts, avoiding duplication. Color-coding the bars by host type further enhances visual clarity, allowing users to quickly distinguish between individual, professional, and other host categories. Sorting the bars in descending order emphasizes the neighborhoods with the highest volume of hosts, making it easier to identify where certain host types are more dominant. This visualization is useful for understanding the competitive landscape, detecting trends in professionalization, and evaluating how hosting styles vary geographically, all of which can inform strategic decisions for both new and existing hosts.

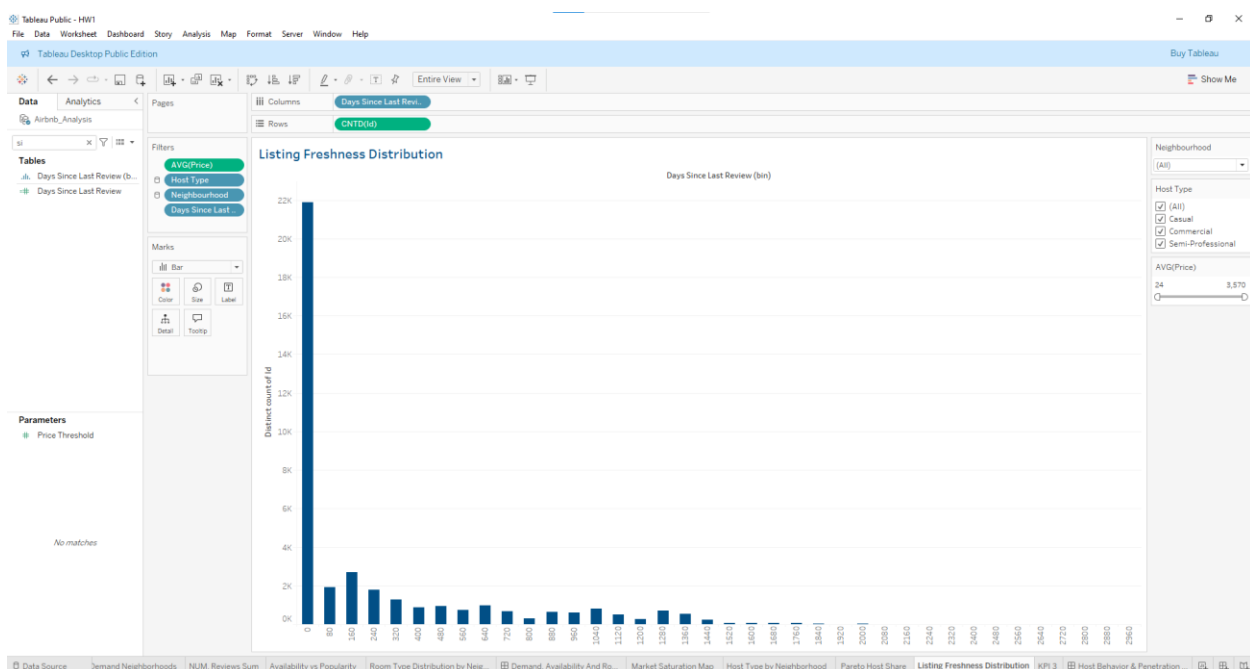


Sheet 3: Pareto Host Share

This visualization illustrates the distribution of listings among hosts, highlighting how much of the market is controlled by top performers. By counting the distinct number of listings (Id) per host and applying a running total calculation converted into a percent of the total, the chart effectively shows the cumulative share of listings as you move down the list of hosts. Sorting the Host Ids in descending order ensures that the most dominant hosts appear first, making it easy to visualize the 80/20 rule or similar patterns, where a small number of hosts control a large portion of the market. This type of chart is particularly useful for identifying market concentration and inequality. It helps reveal whether the short-term rental landscape is dominated by a few large-scale operators or more evenly spread among individual hosts. Such insights are valuable for regulators, researchers, and Airbnb competitors looking to understand platform dynamics and assess the impact of professional hosting on the overall market.



The "Listing Freshness Distribution" chart offers a valuable snapshot of how recently Airbnb listings have been reviewed, serving as a proxy for their current activity status. Displayed as a histogram or bar chart, the x-axis groups listings into bins based on the number of days since their last review (e.g., 0–30, 31–90, 91–180, 180+), while the y-axis represents the number of listings in each category. This visualization helps distinguish between active, recently-reviewed listings and those that may be outdated or inactive. It's an effective way to assess the overall “freshness” of the platform, revealing whether the marketplace is dynamic and engaging or showing signs of stagnation. By highlighting the distribution of listing activity, this chart complements other lifecycle analyses—like host longevity—by shifting the focus to the listings themselves. For potential guests, hosts, and analysts alike, it offers insights into platform health, helping to identify trends in listing turnover and sustained host engagement.



Final Dashboard:

The combined interactive dashboard merges insights from the four previously created visualizations—Market Saturation Map, Host Type by Neighborhood, Pareto Chart of Top Hosts, and Listing Freshness Distribution—into a single cohesive view, allowing for dynamic exploration of New York City’s Airbnb market. With interactive filters for both **Host Type** and **Neighborhood**, users can tailor the entire dashboard to focus on specific segments of the market, revealing nuanced patterns and trends that might otherwise be hidden in the broader dataset.

This unified view not only enhances usability but also strengthens decision-making by linking spatial, categorical, and temporal data together. For instance, users can isolate a particular neighborhood and host type to observe listing density in that area, compare the distribution of different host types, assess how dominant certain hosts are, and determine how recently listings have been reviewed—essentially mapping out both supply concentration and freshness.

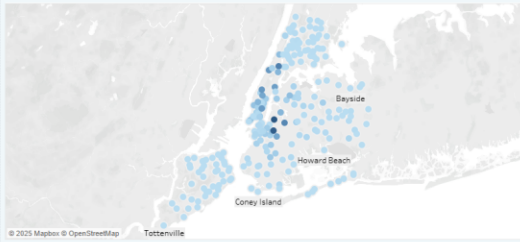
Additionally, two key performance indicators (KPIs) are prominently featured: **average number of reviews per listing** and **average availability over a 365-day period**. These metrics provide further context about listing performance and engagement. A high average number of reviews may indicate trust and activity, while availability reveals how often listings are open for booking—a key insight into host behavior and listing consistency. When combined with the interactive filters, these KPIs enable users to analyze, for example, whether professional hosts in highly saturated neighborhoods maintain more active and frequently booked listings, or if individual hosts in quieter areas still garner strong guest engagement.

Overall, the merged dashboard transforms raw data into an intuitive, story-driven interface that empowers users—from policymakers to business analysts and aspiring hosts—to explore the Airbnb ecosystem from multiple perspectives. It facilitates a deeper understanding of the market’s structure, competition, and liveliness, offering a flexible tool for both high-level strategy and detailed neighborhood-specific insights.

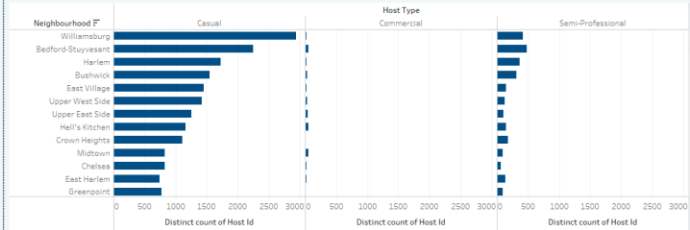
Unlocking the Market: Host & Penetration Trends

Avg. Number Of Reviews
23.27Avg. Availability 365
112.78Distinct count of Id 3,920
Neighbourhood (All)
Host Type (All)

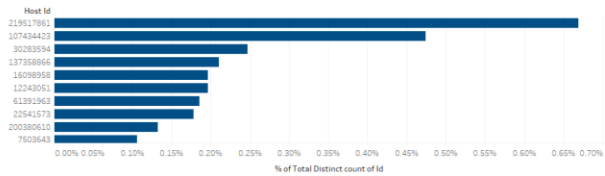
Market Saturation Map



Host Type by Neighborhood



Pareto Host Share



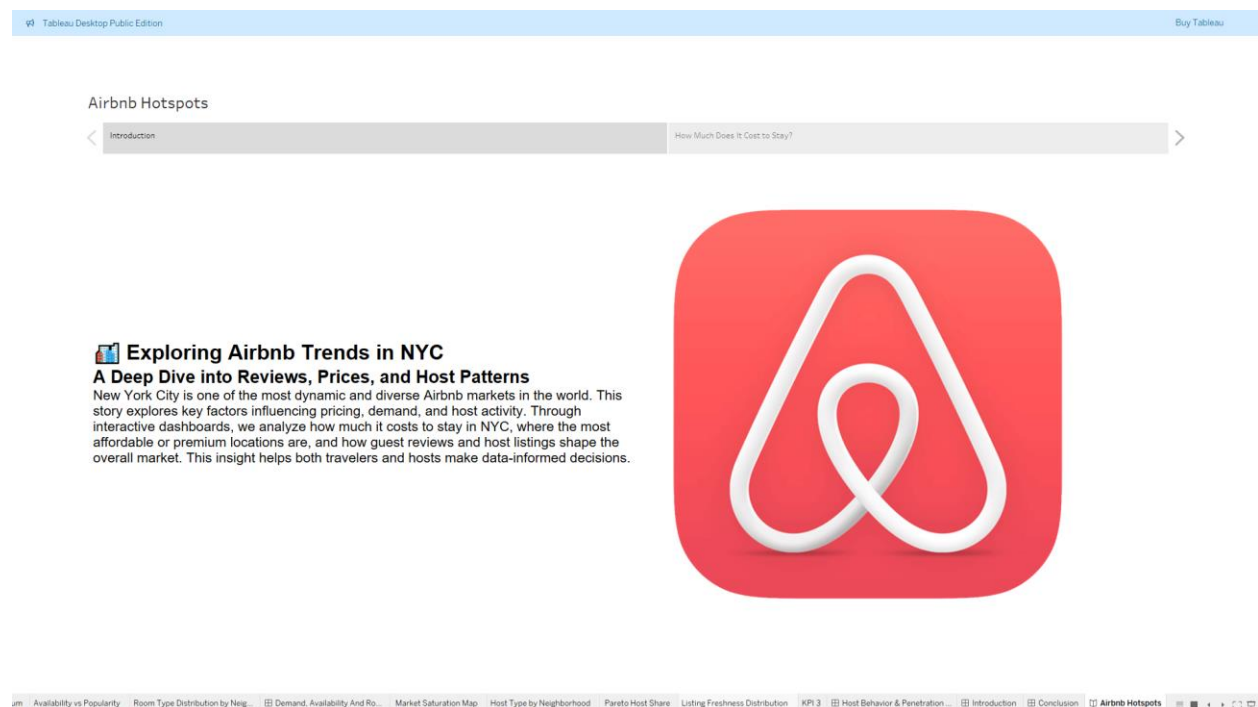
Listing Freshness Distribution



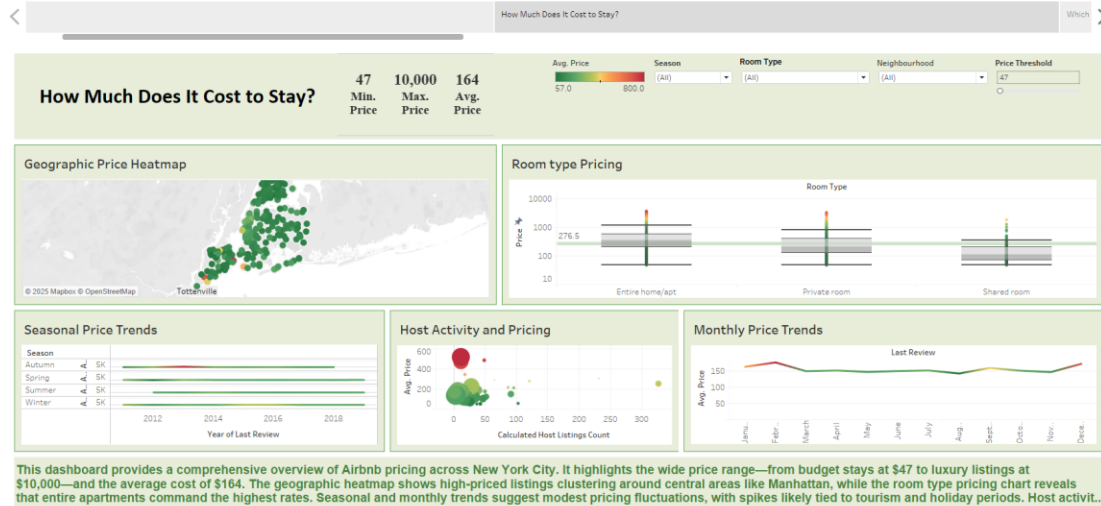
This dashboard reveals that the Airbnb market is characterized by significant concentration and competitive imbalance. A small group of dominant hosts controls a disproportionate share of listings, creating a market where a few players set trends and pricing standards, while many other listings remain underutilized. Additionally, the overall patterns suggest that market saturation is high in key neighborhoods, yet a considerable number of listings show signs of inactivity, pointing to potential seasonal or strategic gaps. These insights highlight the uneven competitive landscape, providing valuable information for hosts seeking to optimize their offerings and for guests looking for more dynamic and responsive rental options.

3. Insights & Storytelling

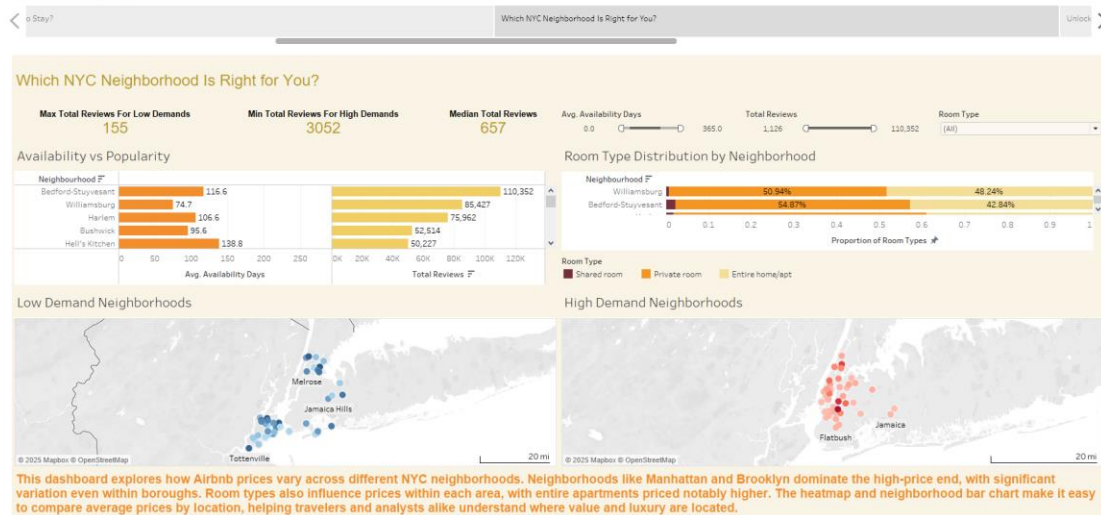
This data story explores key patterns in New York City's Airbnb market, focusing on pricing, neighborhood impact, seasonality, and host behavior. Through interactive dashboards, we discovered that prices vary widely — from as low as \$47 to over \$10,000 per night — with an average around \$164. Entire homes consistently cost more than private or shared rooms, and location plays a major role, with Manhattan listings commanding the highest prices while outer boroughs like Queens and the Bronx offer more affordable options. Interestingly, seasonal changes have only a minor effect on pricing, suggesting that demand remains steady throughout the year. Additionally, hosts managing multiple properties tend to set higher prices, indicating a professional presence in the market. These findings provide useful insights for both travelers aiming to book smarter and hosts looking to optimize their pricing strategies in NYC's competitive short-term rental landscape.



Airbnb Hotspots



Airbnb Hotspots



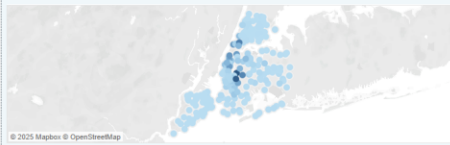
Airbnb Hotspots

< Is Right for You? Unlocking the Market: Host & Penetration Trends >

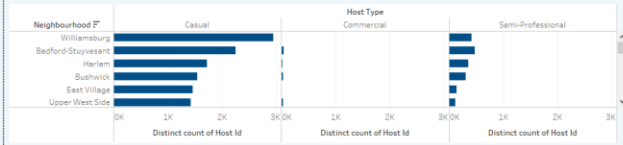
Unlocking the Market: Host & Penetration Trends

Avg. Number Of Reviews 23.27 Avg. Availability 365 112.78
 Distinct count of Id 1 3,920 (All) Neighbourhood (All) Host Type (All)

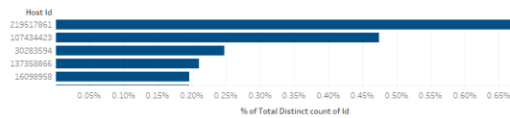
Market Saturation Map



Host Type by Neighborhood



Pareto Host Share



Listing Freshness Distribution



This dashboard reveals that the Airbnb market is characterized by significant concentration and competitive imbalance. A small group of dominant hosts controls a disproportionate share of listings, creating a market where a few players set trends and pricing standards, while many other listings remain underutilized. Additionally, the overall patterns suggest that market saturation is high in key neighborhoods, yet a considerable number of listings show signs of inactivity, pointing to potential seasonal or strategic gaps. These insights highlight the uneven competitive landscape, providing valuable information for hosts seeking to optimize their offerings and for guests looking for more dynamic and responsive rental options.

um Availability vs Popularity Room Type Distribution by Neig... Demand, Availability And Ro... Market Saturation Map Host Type by Neighborhood Pareto Host Share Listing Freshness Distribution KPI 3 Host Behavior & Penetration ... Introduction Conclusion Airbnb Hotspots

Airbnb Hotspots

< Market: Host & Penetration Trends Conclusion >

Key Takeaways: What Drives Airbnb Pricing in NYC?

Pricing Patterns

Prices range from **\$47 to \$10,000**, with an average around **\$164/night**. Entire homes/apartments cost significantly more than private/shared rooms.

Neighborhood Influence

Manhattan listings are consistently more expensive. Outer boroughs like **The Bronx** and **Queens** offer lower-cost options.

Seasonal and Monthly Trends

Prices are **relatively stable** across seasons. Slight dips in early spring and increases in late fall/holiday months.

Host Behavior

Multi-listing hosts tend to charge more. Suggests presence of **professional, business-oriented hosting**.

These insights equip travelers to optimize their booking choices and empower hosts to adapt their pricing strategies in one of the world's most competitive short-term rental markets.



um Availability vs Popularity Room Type Distribution by Neig... Demand, Availability And Ro... Market Saturation Map Host Type by Neighborhood Pareto Host Share Listing Freshness Distribution KPI 3 Host Behavior & Penetration ... Introduction Conclusion Airbnb Hotspots